Ewen Crunkhorn

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EDUCATION

Bachelor of Science, Bioengineering Conc: Biomaterials and Nanomedicine — Graduation: December 2025
George Mason University (GMU)

Masters of Science, Bioengineering — Expected Graduation: December 2026
George Mason University (GMU) — GPA: 3.67

SKILLS, CERTIFICATIONS, & COURSEWORK

Software: Fusion360, Autodesk Inventor, Microsoft Office, Tiamat
Programming Languages: Python, Matrix Laboratory (MATLAB), R
Certifications: Quality Science Professional (Expected Completion: May 2027)
Specialty Equipment and Techniques: Slot Blot, Gels (Agarose, PAGE), Rheology, PCR, AFM, Autosorb, Photolithography, RIE

Laboratory and Technical Experience

Laboratory Research Assistant — June 2025-Present
Veneziano Lab – Institute for Advanced Biomedical Research, GMU
- Conduct literature reviews for advancement of previously developed research pathways.
- Adapt and create SOPS for laboratory use from commercial reagent specifications.
- Create custom DNA nanostructures and visualizations in Tiamat software.
- Perform and assess protein modifications utilizing copper-free click chemistry.
- Conduct agarose and PAGE gel studies for analysis of DNA and protein samples.
- Perform rheological studies of hydrogel samples.

Graduate Student Technician — April 2026 – Present
Nano Fabrication Facility, Center for Biohealth Innovation, GMU
- Operate and maintain specialty fabrication and characterization equipment (photolithography stations, REI, AFM, microscopes, etc).
- Develop and update SOPS, equipment manuals, training materials, and process documentation.
- Assist with inventory management, undergraduate student scheduling, and ordering.
- Train faculty, students, and external users on equipment and provide process engineering support.

ORISE Appointee - Industrial Design Intern, Champaign, IL — June 2024 – August 2024
Construction Engineering Research Laboratory (CERL), US Army Corp of Engineers
- Independently ideated, prototyped, and iterated research platform components.
- Designed, printed, and tested solutions for current platform shortfalls using Autodesk Inventor.
- Ruggedized, shrunk, and improved robotic module form factor and deployment systems.
- Integrated electronics and power systems housing.
- Monitored human-machine interactions and human manipulation of proprioceptive cable-driven parallel robots.

VOLUNTEERING

Election Officer – Officer, Assistant Poll Chief, Poll Chief — 2020 - Present
Fairfax County Office of Elections
- Coordinate with Poll Chief to ensure precinct is correctly set-up and all assigned Election Officials are prepared for election day.
- Ensure the precinct complies with election laws and county codes.
- Instruct assigned Election Officials on precinct best practice and new procedures.
- Manage election officials during 15-hour precinct operation, process Same-Day voter registrations, and coordinate final ballot reporting.

PROJECT MANAGEMENT EXPERIENCE

Project Manager, Senior Design Project — January 2025- December 2025
George Mason University
- Coordinated project scope and timeline with sponsor and faculty advisors, including publication and patent applications post-graduation.
- Managed team assignments, compiled reports, and conducted meetings during development
- Provided hands-on assistance during development with AutoCAD and 3D printing manufacturing
- Oversaw financial and purchasing decisions to ensure budget compliance, found and applied for internal grant to grow budget.
- Won 1st place at department Senior Design Day competition.

Division Director, Member Services — November 2024- Present
Assistant Division Director, Member Services — December 2023-November 2024
Otakorp, Inc
- Manage 8 departments with a staff total of approximately 150 people.
- Coordinate with Division Directors to improve and expand customer service options for 46,000+ members.
- Advise and mentor Department Heads on staff training, retention, and recruitment.
- With Department Heads, develop 5 year plans, reviews yearly SOP updates, and creates yearly budget guidelines.
- Coordinates IT needs and room layout adjustments to ensure service capabilities and member accessibility at convention.
- After 1 year in position, was offered promotion to Division Director.